Setting out to see how to incorporate preference heterogeneity over the product's overall bundle into model via augmenting the U_{ijt} term it is now defined as follows:

$$U_i jt = \sum_j \beta_{ji} X_{jt} + \gamma_i \log(1 + \Xi_{ijt}^2) + \varepsilon_{ijt}$$

$$\text{s.t.}$$

$$\Xi_{ijt} = \sum_j \sqrt{(X_{jt} - \bar{X}_{jt})^2}$$

$$\varepsilon_{ijt} \sim Gumbel(0, 1)$$

This allows for β to vary with the specific characteristics the product holds rather than just a beta for the overall bundle. Variety now takes the euclidean distance of each characteristic of the product's bundle from the amount the consumer has already consumed in the past, allowing for a more nuanced measure of variety that captures the idea that consumers may want to try products that have some different characteristics from their past consumption, but not necessarily products that are completely different from what they have already consumed.

Results reported:

- Utility graphs
- Regression results on log difference in shares
- Inclusive value tables

The regression results make it seem as if γ is not meaningfully accounting for anything missing from the naive baseline regression. The coefficients on the characteristics are not very different across the baseline and LOV regressions, and the coefficient on the variety term is still pretty far from the true value. I wonder if this is a data issue, where maybe I need more variation in Ξ to be able to identify the γ coefficient. I will try to generate some data with more variation in Ξ by upping to 5 consumers, and see if that helps.

1 Utility Graphs

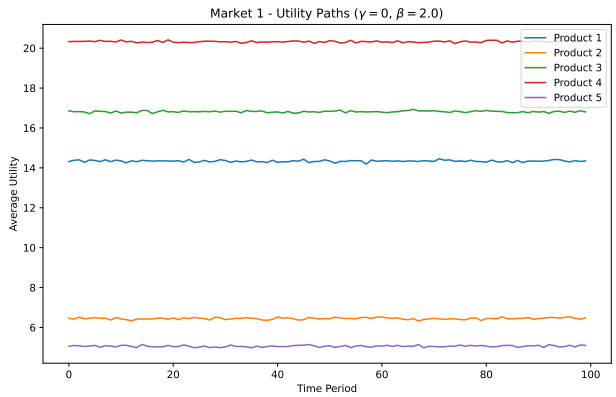

Figure 1: Utility graph for Market 1$\gamma = 0$, $\beta_1 = 2$

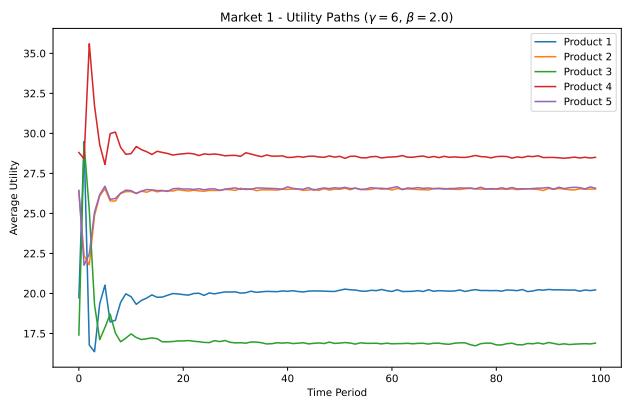

Figure 2: Utility graph for Market 1$\gamma = 6$, $\beta_1 = 2$

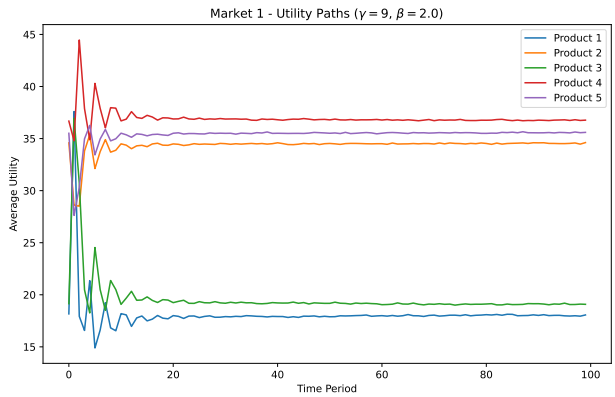

Figure 3: Utility graph for Market $1\gamma = 9$, $\beta_1 = 2$

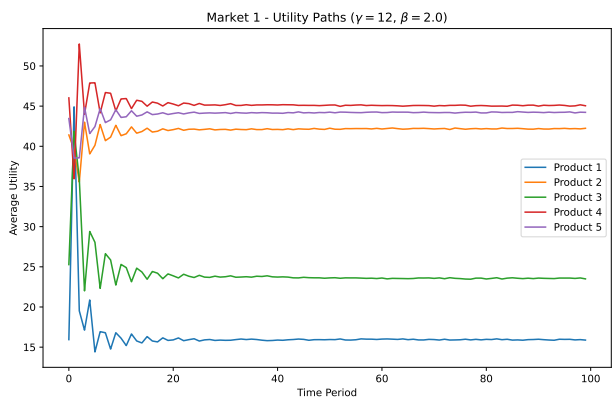

Figure 4: Utility graph for Market $1\gamma = 12$, $\beta_1 = 2$

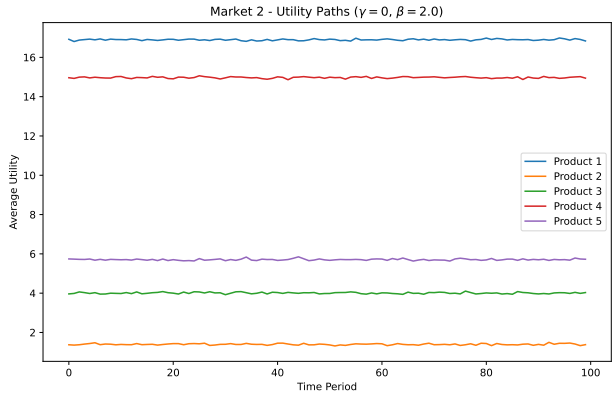

Figure 5: Utility graph for Market $2\gamma = 0$, $\beta_1 = 2$

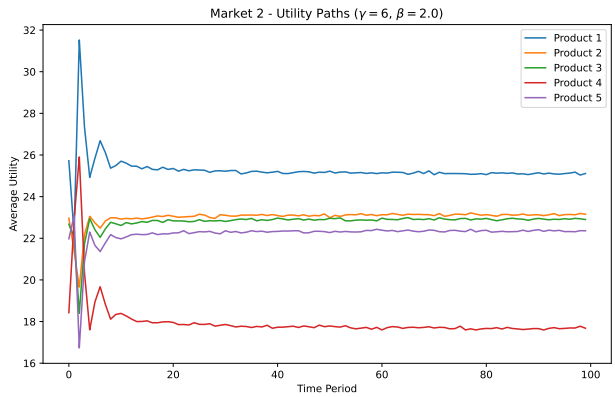

Figure 6: Utility graph for Market $2\gamma = 6$, $\beta_1 = 2$

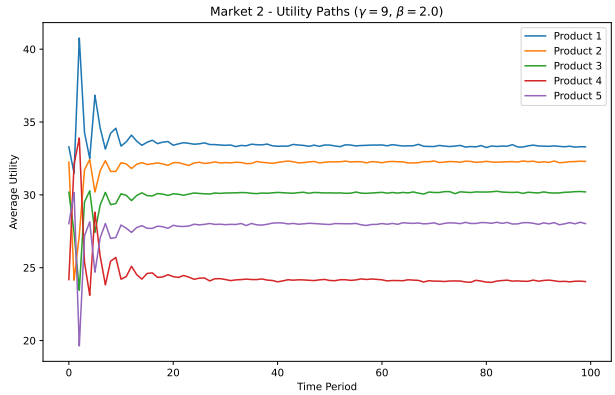

Figure 7: Utility graph for Market $2\gamma = 9$, $\beta_1 = 2$

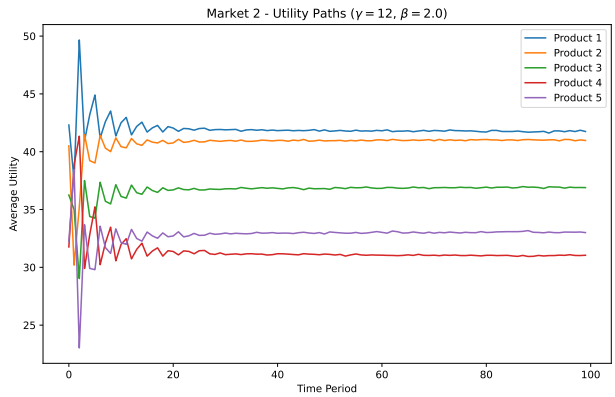

Figure 8: Utility graph for Market $2\gamma = 12$, $\beta_1 = 2$

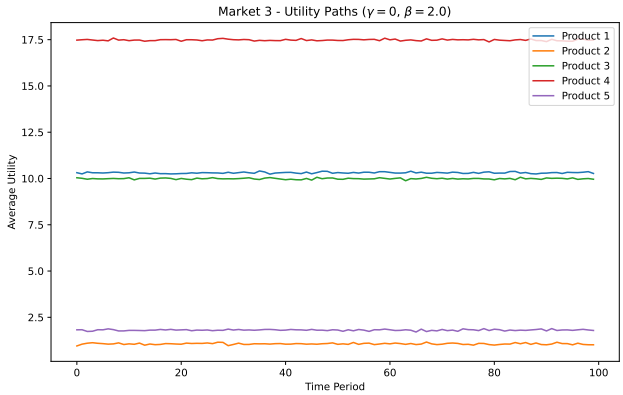

Figure 9: Utility graph for Market $3\gamma = 0$, $\beta_1 = 2$

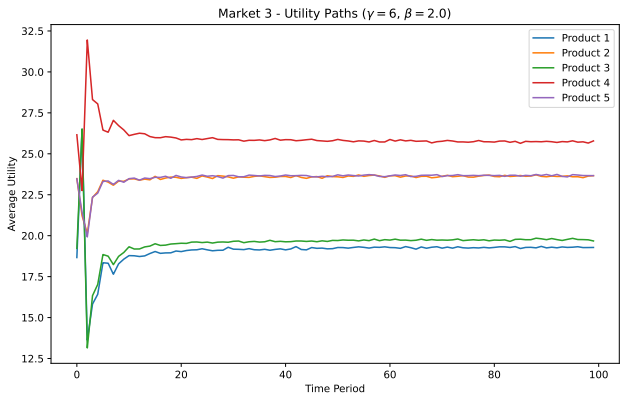

Figure 10: Utility graph for Market $3\gamma = 6$, $\beta_1 = 2$

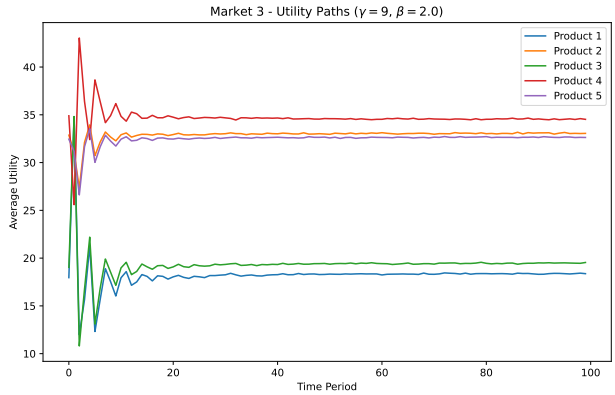

Figure 11: Utility graph for Market $3\gamma = 9$, $\beta_1 = 2$

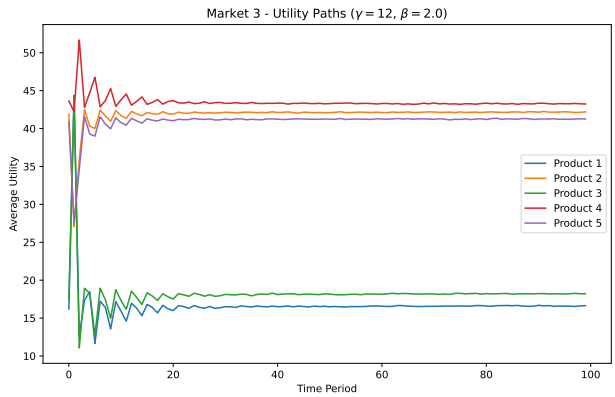

Figure 12: Utility graph for Market $3\gamma = 12$, $\beta_1 = 2$

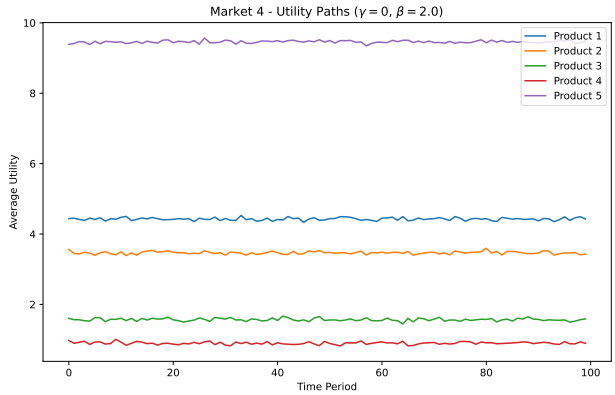

Figure 13: Utility graph for Market $4\gamma = 0$, $\beta_1 = 2$

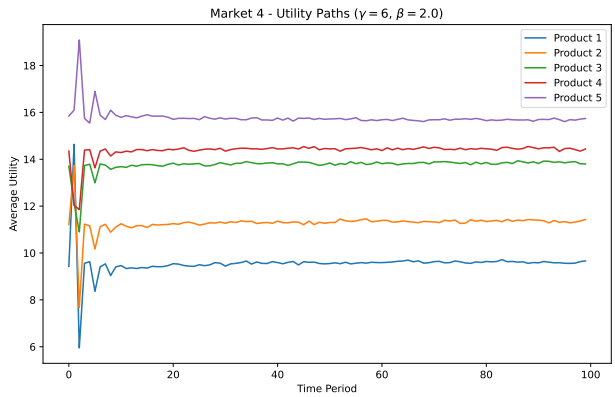

Figure 14: Utility graph for Market $4\gamma = 6$, $\beta_1 = 2$

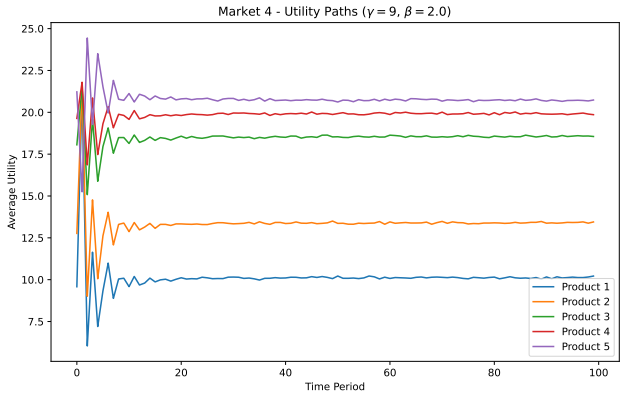

Figure 15: Utility graph for Market $4\gamma = 9$, $\beta_1 = 2$

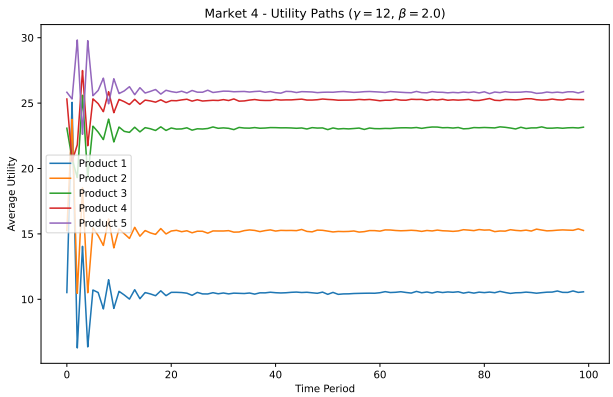

Figure 16: Utility graph for Market $4\gamma = 12$, $\beta_1 = 2$

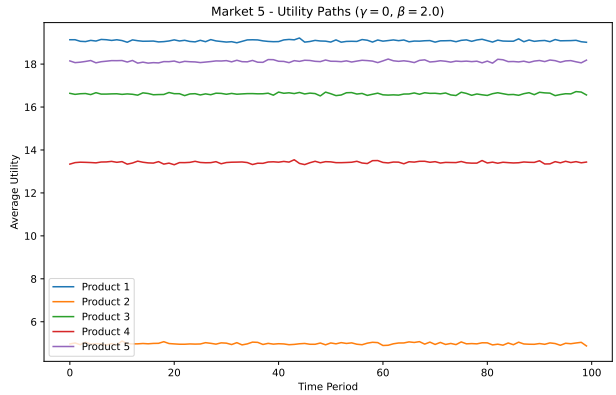

Figure 17: Utility graph for Market $5\gamma = 0$, $\beta_1 = 2$

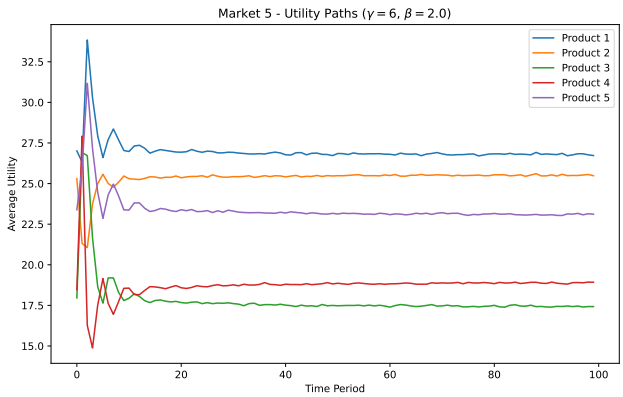

Figure 18: Utility graph for Market $5\gamma = 6$, $\beta_1 = 2$

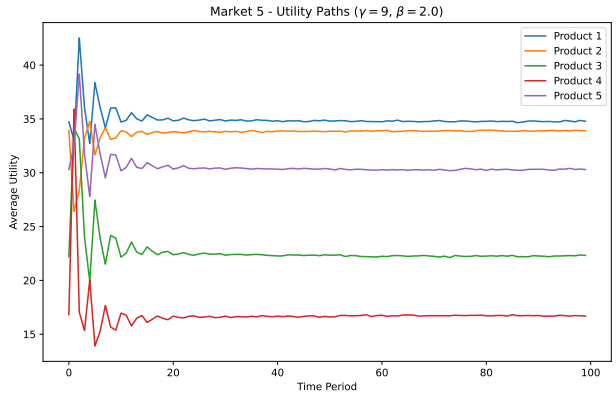

Figure 19: Utility graph for Market $5\gamma = 9$, $\beta_1 = 2$

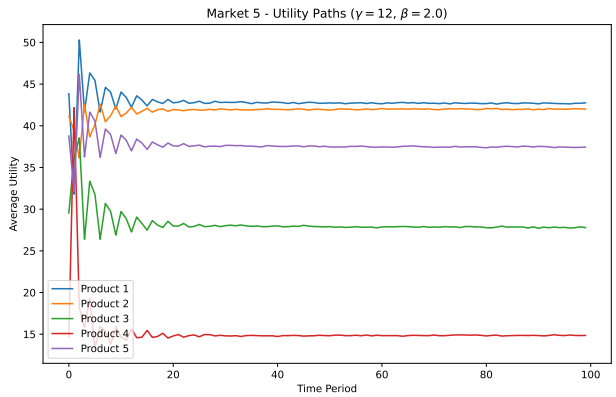

Figure 20: Utility graph for Market $5\gamma = 12$, $\beta_1 = 2$

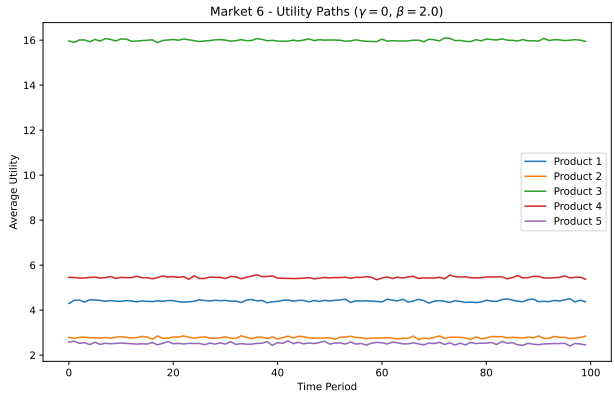

Figure 21: Utility graph for Market 6$\gamma = 0$, $\beta_1 = 2$

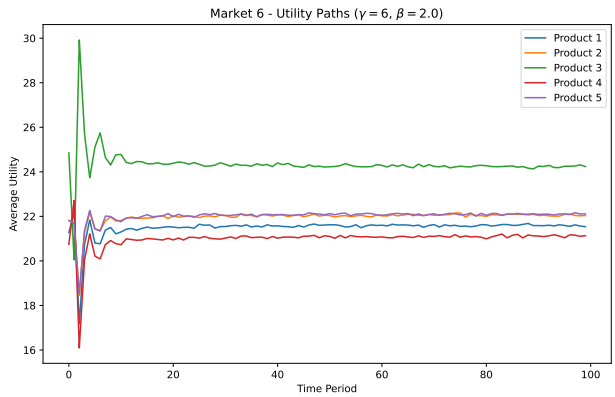

Figure 22: Utility graph for Market 6$\gamma = 6$, $\beta_1 = 2$

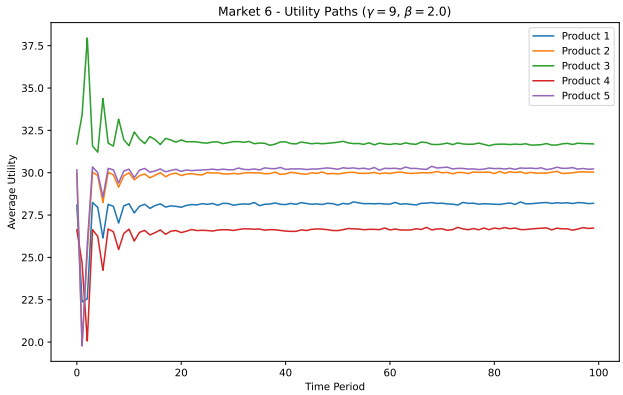

Figure 23: Utility graph for Market 6$\gamma = 9$, $\beta_1 = 2$

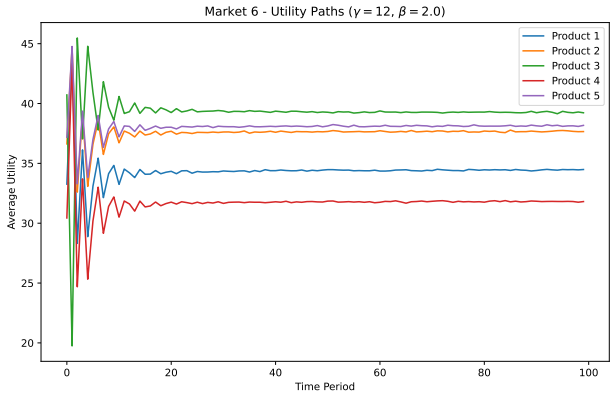

Figure 24: Utility graph for Market 6$\gamma = 12$, $\beta_1 = 2$

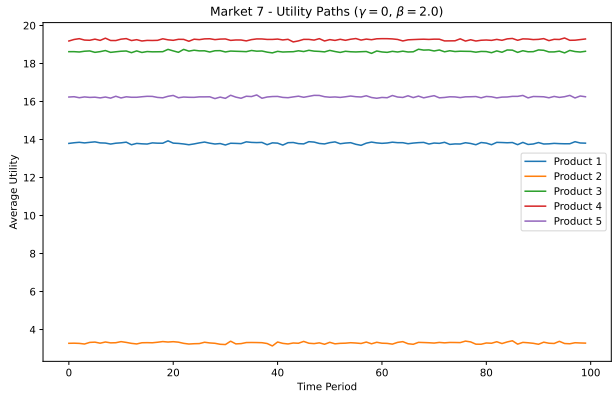

Figure 25: Utility graph for Market 7$\gamma = 0$, $\beta_1 = 2$

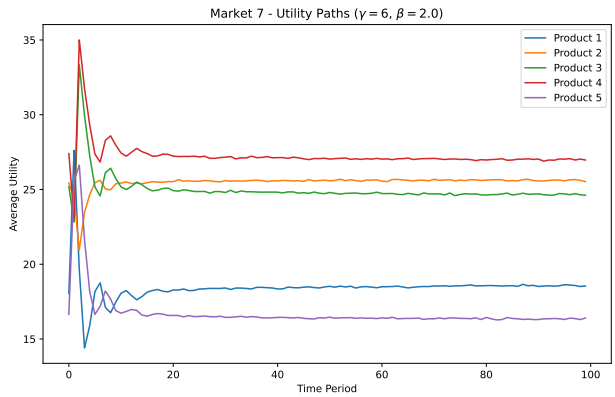

Figure 26: Utility graph for Market 7$\gamma = 6$, $\beta_1 = 2$

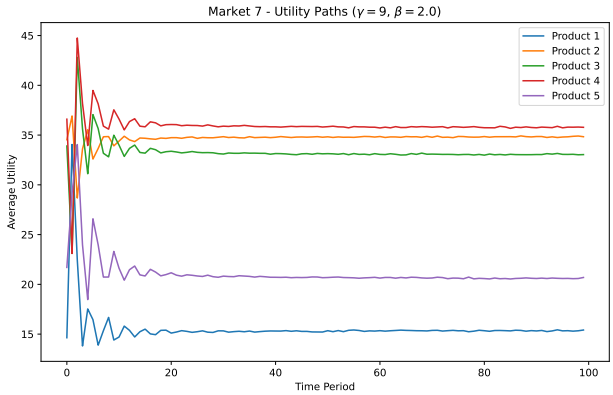

Figure 27: Utility graph for Market $7\gamma = 9$, $\beta_1 = 2$

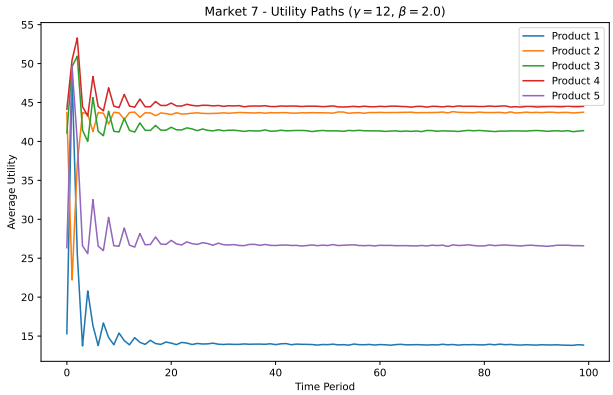

Figure 28: Utility graph for Market $7\gamma = 12$, $\beta_1 = 2$

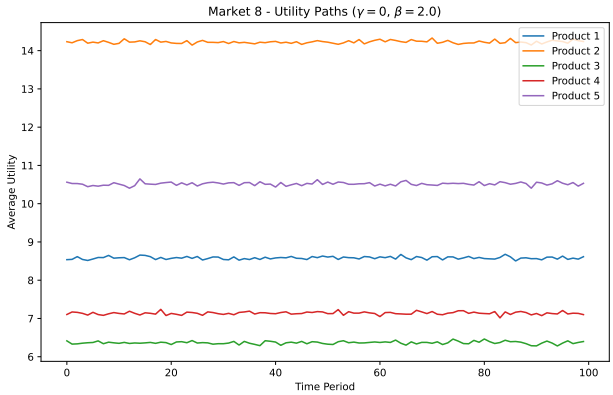

Figure 29: Utility graph for Market 8$\gamma = 0$, $\beta_1 = 2$

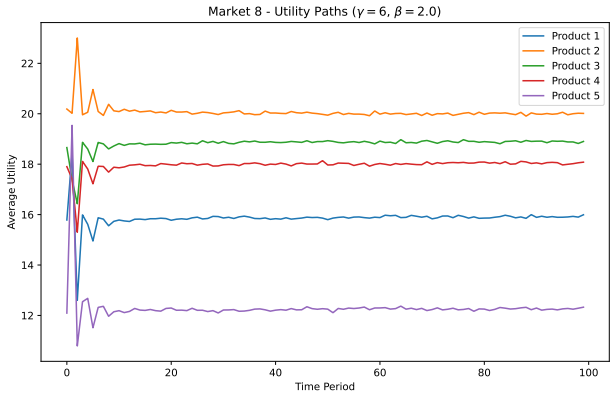

Figure 30: Utility graph for Market 8$\gamma = 6$, $\beta_1 = 2$

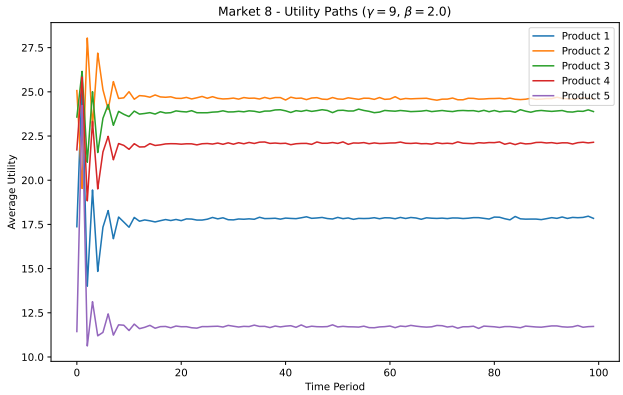

Figure 31: Utility graph for Market 8$\gamma = 9$, $\beta_1 = 2$

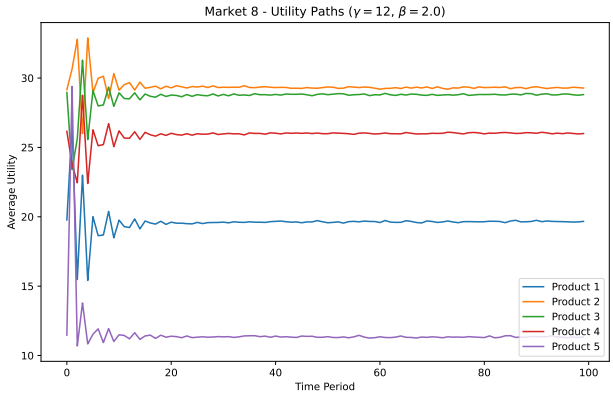

Figure 32: Utility graph for Market 8$\gamma = 12$, $\beta_1 = 2$

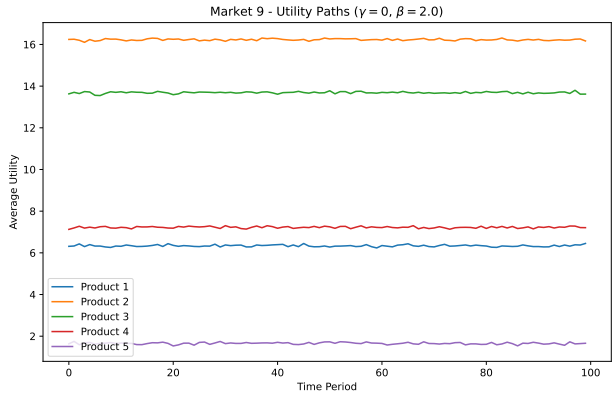

Figure 33: Utility graph for Market 9$\gamma = 0$, $\beta_1 = 2$

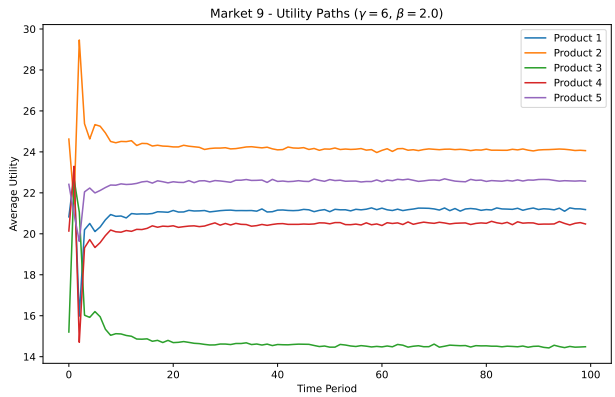

Figure 34: Utility graph for Market 9$\gamma = 6$, $\beta_1 = 2$

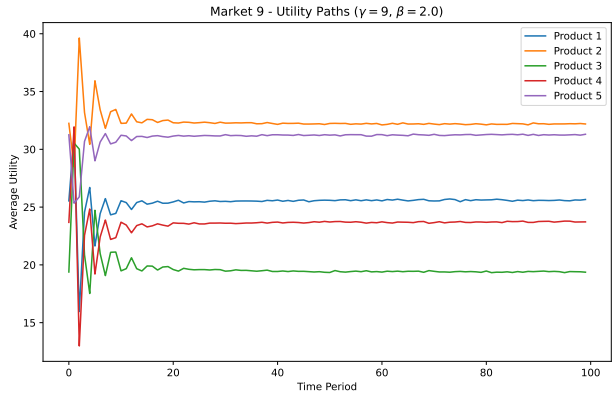

Figure 35: Utility graph for Market $9\gamma = 9$, $\beta_1 = 2$

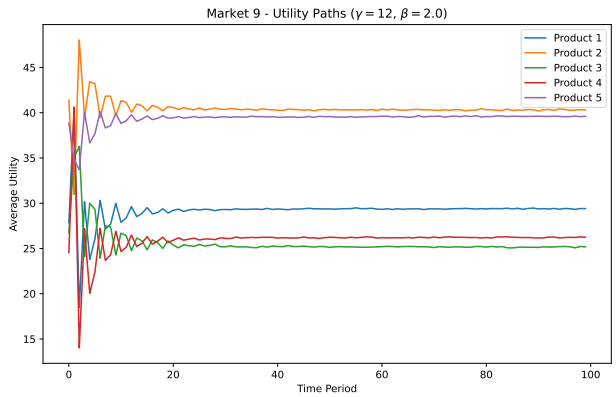

Figure 36: Utility graph for Market $9\gamma = 12$, $\beta_1 = 2$

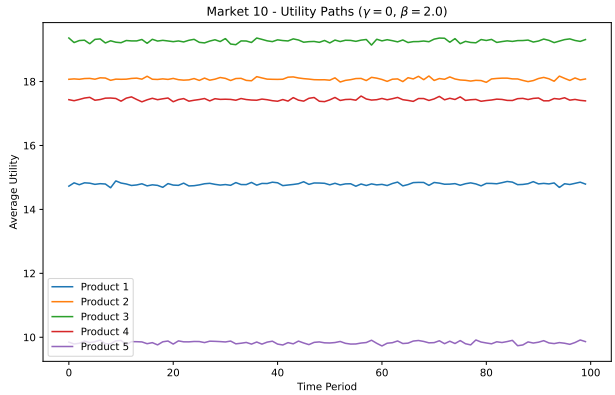

Figure 37: Utility graph for Market 10$\gamma = 0$, $\beta_1 = 2$

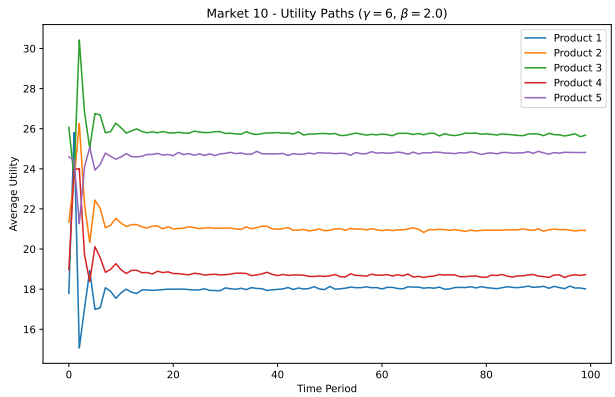

Figure 38: Utility graph for Market 10$\gamma = 6$, $\beta_1 = 2$

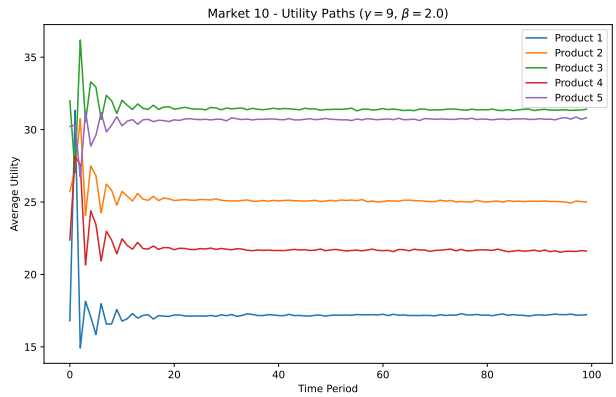

Figure 39: Utility graph for Market $10\gamma = 9$, $\beta_1 = 2$

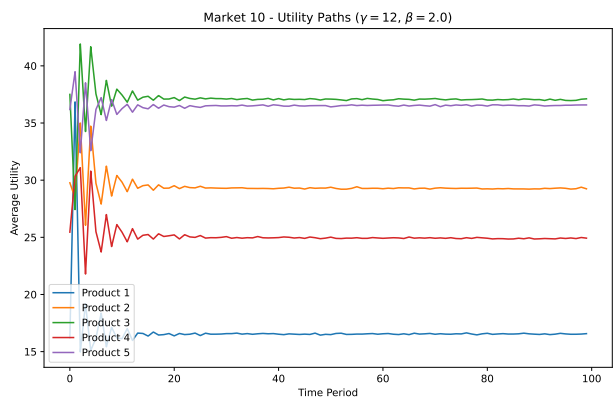

Figure 40: Utility graph for Market $10\gamma = 12$, $\beta_1 = 2$

2 Regression Results on Log Difference in Shares

$\hat{\beta}$	SE($\hat{\beta}$)	R^2
1.7867	0.0035	0.9810

Table 1: Regression results for $y = \beta_j X_j + \varepsilon_j$ with $\beta = 2$, True $\gamma = 0$

$\hat{\beta}$	SE($\hat{\beta}$)	R^2
0.3035	0.0182	0.0529

Table 2: Regression results for $y = \beta_j X_j + \varepsilon_j$ with $\beta = 2$, True $\gamma = 6$

$\hat{\beta}$	SE($\hat{\beta}$)	R^2
0.1414	0.0302	0.0044

Table 3: Regression results for $y = \beta_j X_j + \varepsilon_j$ with $\beta = 2$, True $\gamma = 9$

$\hat{\beta}$	SE($\hat{\beta}$)	R^2
0.0759	0.0439	0.0006

Table 4: Regression results for $y = \beta_j X_j + \varepsilon_j$ with $\beta = 2$, True $\gamma = 12$

$\hat{\beta}$	$\hat{\gamma}$	SE($\hat{\beta}$)	SE($\hat{\gamma}$)	R^2
1.7891	5.2610	0.0077	0.0210	0.9300

Table 5: $y = \beta_j X_j + \gamma \log(1 + \Xi^2)$ with $\gamma = 6$, $\beta = 2$

$\hat{\beta}$	$\hat{\gamma}$	SE($\hat{\beta}$)	SE($\hat{\gamma}$)	R^2
1.8068	8.0897	0.0085	0.0258	0.9519

Table 6: $y = \beta_j X_j + \gamma \log(1 + \Xi^2)$ with $\gamma = 9$, $\beta = 2$

$\hat{\beta}$	$\hat{\gamma}$	SE($\hat{\beta}$)	SE($\hat{\gamma}$)	R^2
1.8415	11.0571	0.0103	0.0323	0.9592

Table 7: $y = \beta_j X_j + \gamma \log(1 + \Xi^2)$ with $\gamma = 12$, $\beta = 2$